

kmaps

P2

ab\cd	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	1	0	1	0
10	1	0	1	0

msop

$$f(a,b,c,d) = a\bar{c}\bar{d} + \bar{a}\bar{c}d + acd + \bar{a}c\bar{d}$$

$$= a(\bar{c}\bar{d} + cd) + \bar{a}(\bar{c}d + c\bar{d})$$

$$\rightarrow A \text{ xor } (C \text{ xor } D)$$

Sum
long

P1

ab\cd	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	0	0	1	1
10	1	1	0	0

msop

$$f(a,b,c,d) = \bar{a}b\bar{c} + a\bar{b}\bar{c} + \bar{a}b\bar{c} + abc$$

$$= a(\bar{b}\bar{c} + bc) + \bar{a}(\bar{b}\bar{c} + bc)$$

$$\rightarrow A \text{ xor } (B \text{ xor } C)$$

P0

ab\cd	00	01	11	10
00	0	1	0	1
01	1	0	1	0
11	1	0	1	0
10	0	1	0	1

msop

$$f(a,b,c,d) = b\bar{c}\bar{d} + \bar{b}\bar{c}d + bcd + \bar{b}c\bar{d}$$

$$= b(\bar{c}\bar{d} + cd) + \bar{b}(\bar{c}d + c\bar{d})$$

$$\rightarrow B \text{ xor } (C \text{ xor } D)$$

$$P2 \rightarrow A \text{ xor } (C \text{ xor } D)$$

$$P1 \rightarrow A \text{ xor } (B \text{ xor } C)$$

$$P0 \rightarrow B \text{ xor } (C \text{ xor } D)$$

xor representation